

AGRI STORAGE

Smart Cold Storage Finder, Reservation Platform & Capacity Synchronization

Document Type: Full Architecture & Code Guide

Event: Silver Oak Kalpvruksh 2.0 Hackathon

Target Audience: Evaluators, Developers, Jury

Stack: PHP 8.x, MySQL (PDO), Vanilla JS, CSS3

Topic: UI Reject Popup Modal & Code Minimisation

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1. Executive Summary & Problem Solved

In India, smallholder farmers lose an estimated **15% to 25%** of horticultural produce (tomatoes, onions, potatoes, carrots, apples) due to inadequate cold chain infrastructure. During harvest gluts, local markets experience severe price crashes. Lacking transparent information on nearby cold storage facilities, farmers are forced into *distress selling* at steep financial losses. Simultaneously, commercial cold storage warehouse owners suffer from underutilized chamber capacities due to offline, fragmented communication channels.

ColdConnect solves this dual problem by providing an integrated, transparent, web-based AgriTech portal that bridges farmers and cold storage operators. Farmers can instantly discover certified facilities, evaluate real-time rates, view temperature capabilities, run live cost estimators, and submit digital reservations. Warehouse managers access a dedicated management portal to monitor capacity, accept reservations with atomic capacity decrements, or reject bookings cleanly using an accessible in-app UI popup modal.

Recent Architecture Improvement

Key Update: Replaced native browser `confirm()` dialog with an accessible, in-app UI Popup Modal for booking rejections. This streamlines code across owner dashboards, eliminates redundant row-level forms, and presents rich booking context (Farmer, Crop, Quantity, Facility) before final cancellation.

2. System Architecture & Technology Stack

ColdConnect is architected following lightweight, robust MVC principles using pure, native web standards without heavy external framework overhead. This ensures instant load times, zero build dependencies, and effortless on-premise or cloud deployment.

Layer	Technology	Architectural Role & Description
Backend Controller & Routing	PHP 8.x	Procedural MVC controllers, strict session RBAC, flash notifications, transaction handling.
Database Layer	MySQL / MariaDB (PDO)	ACID transactions, prepared statements (SQL injection immunity), foreign key cascades, ENUM states.
Frontend Presentation	Pure Vanilla CSS3	Zero external CSS frameworks; custom AgriTech (Emerald) + Ice Cyan tokens, glassmorphism, responsive grid.
Client-side Engine	Vanilla JavaScript (ES6+)	Live cost estimators, date synchronization, event-delegated UI popup modal controllers.
Typography & Icons	FontAwesome 6 + Inter	High contrast visual status badges, intuitive icon cues, clean readability.

3. Relational Database Schema & Data Integrity

The relational database schema is designed for relational consistency and data integrity. It consists of three central entities: users, cold_storages, and bookings.

Entity	Key Fields	Constraints & Integrity Rules
users	id (PK), name, email, password, role, phone, location	email UNIQUE; role ENUM('farmer', 'owner'); passwords hashed via password_hash().
cold_storages	id (PK), owner_id (FK), name, location, total_capacity, available_capacity, temperature, price_per_kg, status	owner_id references users(id) ON DELETE CASCADE; status ENUM('Available', 'Full', 'Maintenance').
bookings	id (PK), farmer_id (FK), storage_id (FK), crop, quantity, start_date, end_date, total_cost, status	status ENUM('Pending', 'Accepted', 'Rejected', 'Completed'); foreign keys cascaded on delete.

4. End-to-End User Workflows

The system coordinates two synchronized user journeys across Farmers and Cold Storage Owners:

A. Farmer Workflow (Search, Match, Estimate, Book)

- **Smart Match Recommendation:** The farmer inputs crop, required volume, and duration. The Smart Match algorithm scores facilities out of 100 based on Proximity (40%), Pricing (30%), and Crop Temperature Suitability (30%), highlighting optimal matches with a 'BEST MATCH' badge.
- **Live Dynamic Estimator:** As the farmer adjusts kilograms or days, client-side JS calculates: $\text{Total Cost} = \text{Quantity (kg)} \times \text{Rate (₹/kg/day)} \times \text{Duration (days)}$ in real-time.
- **Booking Dispatch:** The reservation is created in Pending status, visible instantly on the farmer's booking tracker.

B. Storage Owner Workflow (Review, Accept, Reject, Sync)

- **Real-Time Review:** The owner portal displays pending reservations with farmer contact information, crop specs, dates, and total value.
- **Atomic Acceptance:** If accepted, a PDO transaction verifies capacity, sets status to Accepted, and decrements facility available_capacity. If free space reaches 0, facility status shifts to Full.
- **Modal-Driven Rejection:** Clicking 'Reject' triggers the custom in-app popup modal, showing full reservation summary before confirming rejection. This leaves warehouse capacity unallocated for other farmers.

5. UI Rejection Popup Modal & Code Minimisation

A critical enhancement in ColdConnect 2.1 is replacing the native browser window.confirm() dialog with a modern, animated, in-app UI Popup Modal. This section explains the UX rationale, architectural improvements, and code minimisation achieved.

Why Native Browser Confirm Was Suboptimal:

- 1. Outdated & Unstyled:** Native alerts render OS-specific dialogs that cannot be customized, matching none of the application's visual language.
- 2. Zero Booking Context:** Browser confirm only displays a single line of unformatted text. Owners could not see the farmer's name, crop, quantity, or warehouse.
- 3. Code Redundancy:** Every table row across management pages duplicated entire <form> tags and hidden inputs, inflating HTML DOM size.

Code Minimisation: Before vs After Comparison

Before (Bloated per-row form + inline JS):

```
<div>  
  <button  
    onclick="return confirm('Reject booking #?');">  
    Reject
```

After (Minimised, clean, declarative button trigger):

```
<div data-id=""  
  data-farmer=""  
  data-crop=""  
  data-qty=" kg"  
  data-facility="">  
  Reject
```

Architectural Benefits of Code Minimisation:

- **DRY Principle (Don't Repeat Yourself):** Instead of rendering N separate forms for N rows, a single reusable modal dialog (includes/modal-reject.php) handles all rejections globally.
- **DOM Size Reduction:** Stripped unnecessary form tags and redundant CSRF/action payloads from every row.
- **Centralized Controller:** Client logic in js/script.js utilizes a single event delegation listener, reducing event listener memory footprint and handling dynamic elements seamlessly.
- **Keyboard & Accessibility:** Supports Escape key dismissal, backdrop click dismissal, and accessible ARIA attributes (role='dialog', aria-modal='true').

6. In-Depth File-by-File Code Walkthrough

Comprehensive index of ColdConnect codebase files, their functional responsibilities, and operational mechanics:

File Path	Core Responsibility	Key Functions & Mechanics
includes/db.php	Database Connection	Initializes PDO connection with UTF-8 charset, ERRMODE_EXCEPTION, and FETCH_ASSOC defaults.

includes/auth.php	Session & Security	Role guards (requireFarmer, requireOwner), flash notifications, XSS sanitizer e(), base_url() helper.
includes/modal-reject.php	UI Rejection Modal [NEW]	Reusable popup dialog with warning badge, reservation preview card, and rejection form.
includes/smart-match.php	Matching Engine	Defines crop temperature curves; calculates composite ranking score (Distance, Price, Temperature suitability).
owner/dashboard.php	Owner Overview Portal	Summary metrics, pending bookings table with minimised reject triggers, facility management list.
owner/bookings.php	Reservation Ledger	Full reservations list with status filter (Pending, Accepted, Rejected), farmer details, and UI reject modal.
owner/update-booking.php	Transactional Action Handler	Verifies facility ownership; executes PDO transaction to decrement capacity on accept or reject cleanly.
search.php & booking.php	Farmer Search & Reservation	Facility filtering, Smart Match ranking, date synchronization, and reservation submission.
includes/modal-incident.php	Crop Incident Modal [NEW]	Allows facility owners to log cause of damage (compressor failure, power outage, mold) and calculates guaranteed 85% compensation.
owner/report-incident.php	Incident Action Handler [NEW]	Validates facility ownership and records damage cause, affected crop, damaged quantity, and 85% compensation payout.
includes/header.php	Global Navigation & Bell [NEW]	Notification bell with real-time alert badge, live dropdown reminders for expirations, damage claims, and pending requests.
my-bookings.php	Farmer Reservation Hub [NEW]	Displays remaining days countdown badge (■ X days remaining), 85% insurance coverage badge, and owner damage incident callouts.
js/script.js	Client-Side Interactions	Live cost estimation formula, notification bell dropdown toggle, rejection popup modal, and dynamic 85% incident compensation preview.
css/style.css	Design System & Styling	AgriTech emerald theme, notification dropdown glassmorphism, pulsing countdown badges (.urgent, .active, .expired), and spoilage alert cards.

6B. Agricultural Protection & UX Enhancements

- 1. Owner-Backed 85% Crop Protection Insurance Addon:** Farmers can optionally select Crop Protection Insurance (+5% fee) on the booking page. In the event of cold chamber equipment failure, refrigerant leaks, or power outages, the facility owner is legally committed to compensate 85% of produce value.
- 2. Interactive Leaflet.js Transit Route Map:** Real-time driving route with distance (km) and agro-truck travel time calculated between farmer pickup locations (Ahmedabad, Surat, Vadodara, Rajkot, etc.) and cold storage facilities.
- 3. Remaining Days Countdown Badge:** Live countdown timer badges (■ X days remaining, Ending Soon!, Expires Today) help farmers and owners avoid unintended spoilage or demurrage fees.
- 4. Notification Bell & Live Reminders:** A header notification bell tracks active damage claim compensation, countdown expirations, and pending booking approvals.
- 5. Quantity Flexibility & Validation:** Enforces a minimum of 100 kg with step 1 and maximum capacity bounded across all facilities, eliminating browser step validation errors (e.g. 491/501).

7. Security, Transaction Safety & Edge Cases Handled

ColdConnect is designed with production-grade defenses against common web application vulnerabilities:

- **SQL Injection Prevention:** 100% of database queries utilize PDO prepared statements with parameterized inputs. No raw user input is ever concatenated into SQL strings.
- **Insecure Direct Object References (IDOR):** In owner/update-booking.php, the controller verifies that the target booking facility's owner_id strictly matches \$_SESSION['user_id']. Malicious owners cannot accept or reject other facilities' bookings.
- **Race Condition & Overselling Defense:** Accepting a reservation executes inside a PDO ACID transaction. If warehouse capacity is insufficient (available_capacity < quantity), the transaction aborts with an informative error.
- **Cross-Site Scripting (XSS):** All user-supplied output is passed through htmlspecialchars() via the global e() helper.
- **Double-Action Protection:** Already processed bookings (Accepted or Rejected) cannot be re-processed.

8. Functional Verification & Testing Checklist

The following test matrix validates that all functional and UX requirements are met:

Scenario / Requirement	Action / Step	Observed Behavior	Result
Reject Trigger	Click 'Reject' button in Owner Dashboard or Bookings list	Custom UI Popup Modal animates in smoothly. Browser confirm() does NOT appear.	PASS
Dynamic Details	Inspect modal content	Accurately populates Booking ID, Farmer Name, Crop, Quantity, and Facility.	PASS
Cancel Action	Click 'Keep Booking' or close button (x)	Modal closes immediately. No request submitted; status unchanged.	PASS
Keyboard Dismissal	Press 'Escape' key while modal is active	Modal closes cleanly and restores page scrolling.	PASS
Backdrop Dismissal	Click blurred dark backdrop outside modal dialog	Modal closes smoothly.	PASS
Confirm Rejection	Click 'Confirm Rejection' in modal	Submits POST to update-booking.php; status set to Rejected; flash notification shown.	PASS
Capacity Guard	Check storage capacity after rejection	Warehouse capacity is preserved and NOT decremented.	PASS

Engineering Certification

ColdConnect Prototype Ready for Evaluation: All code minimisations, UI modal improvements, and documentation artifacts have been verified. The system demonstrates an accessible, production-ready solution to agricultural post-harvest preservation and transparent cold chain coordination.